

Value

An Ontology of the Sustainable Residue

0. The Question and the Notation

What is value? This text answers: **what a relation leaves behind, when that relation has a real cost and that cost remains sustainable.**

For this answer not to remain a mere formula, it needs a stable object that can be followed from beginning to end. Let A and B be two terms that compose — through an exchange, a bond, a perception, an act. We denote by $R(A, B)$ what this composition leaves behind: neither A , nor B , nor their sum, but a third object, produced by their encounter, which existed in neither of them separately.

Everything that follows is a single question asked twelve times at different levels: **what becomes of R ?** Section by section, we ask under what condition R exists, what it structurally is, where it reappears under other names, how it accumulates, what it becomes in each domain of value, what derails it, what it prescribes, how far it extends, what would refute it, and what it can never say about itself. No section introduces a new concept without explicitly tying it back to R ; none relies on outside authority to advance the argument.

This imposes a discipline of presentation. This text owes confirmations to thinkers from very diverse fields — Whitehead, Simondon, Hegel, Marx, Kant, Shannon, among others — which it would be dishonest to pass over in silence. But a confirmation is not a demonstration, and mixing the two blurs the difference. They are therefore grouped, section by section, in a separate and clearly marked paragraph — a **Reference Point** — which never carries the weight of the argument and can be skipped without loss by anyone who wants to follow only the line of R .

I. From Substance to Relation

Where does A begin, before any composition?

The ordinary answer is: in itself. A thing would first be what it is, and would only afterward, accidentally, enter into relation with others. This answer presupposes that A can be designated independently of any $R(A, -)$ — independently of any trace it carries of its past relations. But this is precisely what no procedure allows one to do: to describe A is always to enumerate what it relates to, never to reach a core that would remain once everything else has been stripped away.

The hypothesis adopted here is therefore the reverse: A is not primary; it is itself a stabilized residue — an R dense enough and slow enough to dissipate that it behaves, provisionally, like a term. There is no term that precedes relation; there are only residues thick enough that we treat them as if they were terms.

This hypothesis cannot be proven in absolute terms — one cannot stand outside all relation to verify that no residue of substantial excess exists. It is justified pragmatically: it is the only one that allows everything that follows in this text to be reconstructed without ever positing a first term by decree. It is a working program, not a metaphysical verdict — and its true test will only come in section XI, when we ask what could refute it.

An apparent tension must be resolved before going further, because it returns every time we speak of one node rather than another. No scale is privileged in the relational universe: nothing designates, in absolute terms, the position from which everything else should be measured. And yet every node *in fact* occupies the position from which it measures — for A , it is always from A that $R(A, -)$ is computed. The two statements stop contradicting each other as soon as we specify for whom each is true: the first holds absolutely, the second holds indexically, for each node relative to itself. This is not a metaphysical privilege granted to certain nodes — it is what it structurally means to occupy a position within a network of relations. The Yoneda lemma carries exactly this answer: $h^A = \text{Hom}(A, -)$, the collection of all of A 's relations to everything else, by itself exhausts the identity of A ; each object is therefore, formally, its own total perspective on everything else — a harmless fact rather than a privileging one, because it holds symmetrically for every object in the category, without exception.

This indexicality does not hold only for position — it holds for time as well, and it must be said here rather than left implicit until section XII. Nothing designates, in absolute terms, the instant from which everything else should be dated — and yet every node *in fact* lives from its own present, the one that its accumulated R gives it. A shared present, when it appears, is therefore never a single instant imposed from outside on the nodes it would date: it is an interval that narrows as sufficiently independent compositions come to constrain it from several sides at once, without ever reducing to a point fixed in advance by a clock external to them. Two separate nodes can thus each compose their own duration without either needing to wait for the other for its duration to count as real — which directly sets up the question section XII will ask about encounter.

Reference Point. Whitehead makes the *actual entity* a node of prehensions rather than a qualified substance; Rovelli shows that a physical state has a determinate value only relative to an observing system. Simondon inverts the same hierarchy in the field of individuation: the individual is never primary, it is a phase of the process, a provisionally stable R . The formula “each node is its own total perspective on the rest” has an older formulation in Leibniz — the monad as a living mirror of the universe — and in the Buddhist metaphor of Indra’s net: where these two images require a pre-established harmony or remain a religious figure, the Yoneda lemma demonstrates the same result within category theory, without any transcendent postulate.

II. The Condition: Irreversibility

Does every composition leave a nonzero R ?

No. If the composition of A and B can be undone without any trace remaining — if the final state is strictly indistinguishable from the initial state — then $R(A, B) = 0$. Such a composition has produced nothing: with respect to its effects, it is indistinguishable from the absence of relation. This is not a rare case of an unproductive relation; it is what it structurally *means* for a relation not to have taken place in the sense that matters.

$R(A, B) \neq 0$ **therefore requires that the composition be irreversible**: that undoing it be itself a new composition, in turn leaving its own trace. This is not a contingent property that some relations have and others lack; it is the minimal condition for R to exist.

This condition delimits an interval. At the lower limit, *transparency*: $R = 0$, nothing remained. At the upper limit, *destruction*: the composition was so intense that it destroyed A or B as possible carriers of a future R — something took place, but it ruined its own possibility of repeating itself. It is between these two limits, and only there, that R can exist *and* accumulate. We will meet this interval again under its proper name — sustainability — in section VII; it appears here for the first time, in its barest form.

One confusion must be guarded against, however, because it would let back in through the window what this section has just excluded through the door. Irreversibility is a condition of R 's *existence*; it is not, by itself, a condition of *accumulation*. A trace can be perfectly irreversible — impossible to undo without in turn leaving a trace — and yet sterile: incapable of entering into a future composition, frozen out of any possible uptake. A scar is irreversible; it orients no future composition, it merely persists. $R \neq 0$ is therefore not enough to guarantee that R counts in the full sense of the term; it must, in addition, remain *mobilizable* — capable of being taken up by a later composition without having to be remade from nothing. We will call this second requirement, distinct from the first, the condition of **uptake**. Section III will show that it is not an arbitrary addition but a direct consequence of what III calls the *orienting* property of R ; section VIII, at its close, will make it the object of its own diagnosis — the case where irreversibility is achieved but uptake never is.

Reference Point. The analogy with the second law of thermodynamics is illuminating but limited: entropy concerns countable microscopic states, whereas the irreversibility defended here is a thesis about the logical structure of every relation, affective as much as physical — two structurally analogous irreversibilities, not an instance and its law. Kit Fine showed that the grounding relation is irreducibly asymmetric without giving its origin; this section supplies it: if A grounds B , undoing this composition would be another one, not the first — hence the asymmetry.

III. What R Is: The Residue as Cross Term

From what operation does R proceed — and where should it be sought?

Two answers are possible. The first makes R a remainder by *subtraction*: one starts from a whole, removes a part, and what remains was already contained in the whole — it would suffice to dig deeply enough into A or B to find it there. The second makes it a *cross term*: in $(A + B)^2 = A^2 + B^2 + 2AB$, the term $2AB$ was contained neither in A nor in B taken in isolation; it did not pre-exist, it is a product of their encounter.

It is this second sense that this text adopts for R . The consequence is immediate and governs the rest of the text: seeking the origin of $R(A, B)$ in A alone or in B alone is not difficult — it is ill-posed. There is nothing to find there, not because it would be lost, but because R never resided there.

R thus defined has four properties that chain together rather than simply adding up. It is *irreducible* (III, above) — from which it follows that it can be *persistent*: in any system that does not entirely collapse, no internal operation can bring R back to zero without collapsing the system itself. Because it persists, it can *accumulate* from one composition to the next rather than dissipating at each cycle. And because this accumulation is *non-commutative* — $R(A, B)$ combined with a third term does not give the same result depending on the order — R becomes *orienting*: a system that carries an R does not meet its future compositions from nothing; it meets them from what R has already accumulated, which serves as structural memory. This last property is the bridge to section IX. It is also what answers, in advance, the condition of uptake posed in II: an orienting R is by construction an R that has been taken up — otherwise it would orient nothing.

A second non-commutativity, distinct from the first, must be made explicit rather than left implicit in the notation $R(A, B)$. Nothing in the definition of the cross term requires that the encounter affect both terms it composes equally. Writing $R(A, B)$ suggests a symmetry that nothing guarantees; strictly speaking one must distinguish $R_A(A, B)$, what the composition leaves *on the side of A* , from $R_B(A, B)$, what it leaves *on the side of B* — and must not presuppose that $R_A(A, B) = R_B(A, B)$, nor even that their sum is conserved from one encounter to the next. An encounter can enrich A without enriching B to the same degree, without this signaling theft or extraction: this is simply what it means to compose asymmetrically. Treating $R(A, B)$ as a single, symmetric object, as the notation does for convenience from section 0 onward, is a legitimate simplification as long as no question bears on the distribution; as soon as such a question arises — and it will, in VII, regarding the affective, and in VIII, regarding pathology — the notation must split in two.

Reference Point. Shannon's mutual information, $I(A; B) = H(A) + H(B) - H(A, B)$, is a case where R is rigorously computable as an exact cross term: it equals zero if and only if A and B are statistically independent. This establishes that *at least one* formal domain makes R computable — not that every R , economic or affective, has this precise nature. The extension to other domains remains an analogical use, a point section XI will return to directly.

IV. First Return: R in the Oldest Dialectic

Is R merely an artifact of this text, or does it recover, unprompted, an already-known structure?

Test: take the oldest logical structure in Western philosophy and check whether R reappears there under another name. Hegel opens the *Science of Logic* with the claim that pure Being and pure Nothing have the same content — no determination distinguishes them — and yet occupy opposite positions. Their common truth is not a third thing added on; it is Becoming: the very tension that holds them together and makes each pass into the other.

Put in the vocabulary of this text, this structure reads directly: Becoming is R (Being, Nothing). It was contained neither in Being alone nor in Nothing alone — exactly the cross term defined in section III — and it is not a third term but the effect of their composition.

This reading allows us to distinguish two words ordinarily conflated. **Actuality** is the set of everything effectively composed — things, bodies, stabilized states: it is R sedimented, the accumulation of every count-for-one already produced. **Reality** is the movement itself, of which actuality is only the residue — the passage between Being and Nothing, not its deposit. Naming actuality “reality” would reproduce, at the summit of this text, the very error it has charged the metaphysics of substance with since section I: taking the stabilized residue for the foundation rather than for what it is, an effect.

Reference Point. Lawvere showed that the scheme of *adjunction* — two functors pointing in opposite directions, linked by reciprocal natural transformations — is the mathematical form of this structure: not a translation of Hegel, but the independent discovery of the same invariant. Badiou makes the empty set the only proper name of being, and insists that the One that results from it is never anything but an effect — the “count-for-one” — never a primary Unity. In the vocabulary of this text, this count-for-one is an R : the local, provisional trace of an act of composition that retains something rather than falling back into the indifference of the void.

V. Second Return: R Accumulates

Once produced, what becomes of R over time?

It does not remain equal to itself. An R produced at instant t undergoes two opposing forces: a tendency to maintain itself as is (its *elastic* component) and a tendency to dissipate or transform under the pressure of new compositions (its *viscous* component). The ratio between these two components determines what can be called R 's *relaxation time* — how long it continues to orient future compositions before dissolving into noise.

Two modes of accumulation must be distinguished, since conflating them would be fatal. *Synchronic* accumulation is the density of relations a node concentrates at a

given instant — legible in an instantaneous diagram, but which says nothing about R properly speaking, since R exists only across the duration of the cycle that produces it. *Diachronic* accumulation is the enrichment of R through the repetition of the cycle over time: it is this, and this alone, that produces what can be called a **corpus** — the sediment of all past compositions, which orients those to come without prescribing them.

This distinction has a consequence that must be made explicit, because it is easy to miss once one looks at only a single instant at a time. At any moment, a node simultaneously carries two quantities of different natures, and conflating them amounts precisely to conflating the synchronic and diachronic accumulation just distinguished. The first resets to zero at every new cycle of composition: it is what is currently mobilizable, the current state of the elastic stock defined in VI, measured since the last event that affected it. The second never resets to zero: it is the total of what the corpus has sedimented, cycle after cycle, since the node's first non-trivial composition. A system can therefore go through a cycle in which the first quantity falls to nothing — a loss, a rupture, a silence — without the second being affected by it: the local reset does not erase the sedimentation that precedes it. It is this distinction, more than any other, that prevents confusing a momentarily impoverished system with a system that simply never accumulated anything.

Three laws govern this diachronic accumulation. **Repetition:** an R produced by a single composition is fragile; repeated in similar contexts, by many nodes over many cycles, it generates a stable, transmissible association — the very mechanism of sedimentation. **Variation:** an R that repeats without ever varying tends toward collapse, in the sense of III — its cycles converge toward identity, and pure repetition ends up undoing what it was stabilizing. A tension between repetition and variation is therefore necessary: too much of one freezes the corpus, too much of the other prevents it from forming. **Selective transmission:** not all R propagate equally; those that propagate best simultaneously satisfy three criteria — memorability, usefulness in situations actually encountered, and valorization by nodes with high relational density. This third criterion introduces a structural asymmetry: the R 's of the densest nodes disproportionately structure the shared corpus. This is not a pathology in the sense of VIII — it is the normal structure of any value system, and it predicts, without any malice being necessary, that intellectual corpora tend to over-represent the R 's of those who already controlled the channels of diffusion.

The densest medium of this accumulation is speech. A word is never a beginning: it carries, compressed, the trace of everything the corpus has already composed before it, and it orients — without prescribing — what the next speaker will be able to draw from it. Speech is itself an R , subject to the three laws above; the great civilizational corpora — mythologies, legal codes, sacred texts — are not prescriptions handed down onto a practice that had been waiting for them, but sedimented residues of a practice already underway, which in turn orient what comes after them.

Reference Point. David Graeber, in his genealogy of debt, advances a compatible thesis: money is born from debt relations, not the other way around — currency would be an institutionalized R rather than a starting point. A limit must nonetheless be maintained: that the formal structure is correct does not establish that some particular historical sequence is its cor-

rect causal explanation. The contemporaneity of two phenomena remains a structural parallel, not proof of mechanism.

VI. The Vocabulary of R : Elasticity, Viscosity, Liquidity

How can one speak precisely of an R that behaves at times like a solid, at times like a fluid?

The vocabulary of viscoelasticity, borrowed from materials physics (Zener), gives a quantitative grip on the properties already named in III and V, without changing their nature. An elastic material stores and fully restitutes; a viscous material dissipates irreversibly; a viscoelastic material does both, according to a characteristic ratio and a **relaxation time** — the time it takes a perturbed system to return to equilibrium. Applied to R : its elastic component is what it stores — the memory of the composition that produced it; its viscous component is what it dissipates — the part of the composition that fades over time.

One must be precise about the status of this vocabulary: it is an *applied* model, chosen for its pedagogical clarity, not deduced from the argument of sections I through IV. A competing vocabulary — thermodynamic, cybernetic, purely categorical — could in principle account for the same structure. What the model establishes with certainty — that R has a component that persists and a component that dissipates, in a proportion that varies by system — was already true independently of any materials physics; what Zener adds to it is a convenient way of saying so, not a necessity.

This model allows us to specify a **causal chain**, which is the deep structure of value and which must be stated in the exact order in which it occurs, because it is this order that section VIII will show inverted in every pathology. **Elasticity** comes first: the capacity of a system to store R rather than dissipate it immediately. **Velocity** follows from it: the speed at which R , once stored, circulates and propagates. **Liquidity** measures the accessibility of R : the extent to which it can be mobilized in new compositions. **Utility** is the capacity of R , thus rendered mobilizable, to effectively orient new compositions — the orienting property of III, now quantified. **Value** is the terminal emergence of this chain: what R has become once these four conditions are simultaneously satisfied.

Value is therefore not the premise of the chain but its endpoint. It is the fundamental error of every ontology of value to have inverted this — to start from value as an immediate given (a price, a signal) and derive its conditions after the fact, when it is only the outcome of a process that begins in elasticity and ends only at the final step.

Fusion. This vocabulary allows us to name a case that III and V did not yet distinguish: the case where A , predominantly elastic, and B , predominantly viscous, compose in a way that does not merely produce one more $R(A, B)$, but transforms the terms themselves. Two ways such a composition can unfold must be distinguished, because only one of the two actually produces something new.

In the first, A cedes its viscous part to B and receives in return B 's elastic part: each comes away specialized, one-sided, but separable. This is not a fusion — it is

a redistributed inventory. The result always contains, in embryo, the trace of who brought what; one could in principle undo the operation and recover A and B in their original composition. Such a result falls back under the ordinary definition of III: two separable terms, a sum, never a genuine R .

In the second, the surrender is total and bilateral in both components at once: A 's viscous part and B 's do not remain two juxtaposed quantities — they become a single viscosity, of which one can no longer say what fraction comes from which; the same holds for the elastic part. Only this non-separability satisfies the condition set out in III for any genuine $R(A, B)$ — neither in A , nor in B , nor in their sum, not even by after-the-fact reconstruction. The result is then a higher-level entity: an R that carries a full elasticity and a full viscosity simultaneously, rather than the single dominant trait each of the two terms carried separately — and which, by IX, orients its environment with a reach no one-sided R has, able to guide through its stable part as much as through its mobile part.

This condition of totality admits no intermediate degree. A partial surrender — one of the two terms retaining, in one component or the other, a share to guarantee itself an exit — does not produce a weakened fusion; it produces nothing of that kind. What remains is a coexistence that displays the outward signs of the bond without satisfying its condition — structurally akin to the capture that VIII will describe further on, where the form of the phenomenon is present without its condition of production being met. From this requirement follow two real regimes, not two options to choose between a priori: if the surrender is total, the higher entity necessarily forms; if it is not, the composition falls back into the ordinary case, with A and B remaining separable and free to compose elsewhere, and the closure of XII applies without modification. Separation is therefore not the alternative to fusion; it is simply what happens by default whenever totality is lacking.

One last clarification, which avoids an easy confusion. The asymmetry $R_A(A, B) \neq R_B(A, B)$, established in III for the general case, does not survive inside a successful fusion: once the viscosity and the elasticity have each been fused into one, there is no longer any “what A keeps” distinct from “what B keeps” within the formed entity. This asymmetry therefore describes only the trajectory that made the fusion possible — the dissimilar contributions of A and B that gave their composition a reason to occur — not a division that would survive within it. This is a direct consequence of non-separability itself: what is truly fused ceases, by definition, to retain the trace of who brought what.

VII. Third Return: R in the Four Domains

What becomes of R once we stop treating it abstractly and ask: are economic, affective, aesthetic, and moral value the same R ?

Yes, except that none is reducible to the other three — because what is common is the formal structure of R , not its content, and the four domains instantiate this structure with radically different relaxation times.

Economic. R is the surplus a trade produces beyond what each party brings — Marxian surplus value is one explicit theory of it. An economic R ceases to be sustainable when its extraction ruins the relation's capacity to reproduce itself; institutions (contract, currency, market) are the devices that keep R within the interval defined in section II, between transparency and destruction.

Affective. R is what two people continue to carry of each other — dependencies, obligations, shared history — stored elastically by the bond. Its size is not the criterion: an affective R that exceeds sustainability produces oppression, not value. The criterion is that R remains freely carried, at a velocity the parties to the relation determine themselves rather than suffer.

Aesthetic. A work is an object whose R has an exceptionally long relaxation time — potentially unbounded. Each contact with the work produces a new R that feeds future encounters without ever exhausting the source; this is the exact description of what Kant calls an aesthetic idea, a representation that gives rise to thought without any concept ever exhausting it.

Moral. A commitment is a relation whose elastic component of R is voluntarily maintained against the viscous temptation of immediate dissipation. To keep one's word is to refuse to let R slacken into the opportunism of the moment — what the categorical imperative formulates as structure rather than as content.

Attempts to reduce these four R 's to a single measure (price, for instance, as the measure of all value) necessarily fail: they homogenize heterogeneous relaxation times. It is this error, rather than a moral failing, that the following section diagnoses.

VIII. When R Goes Wrong: Pathologies

Can one of the four variables of section VI go wrong without the others following — and what happens if it does?

The characteristic pathology of every value system is to **select directly on the apparent trace of R** — the price, the signal, the indicator — rather than on the conditions that produce it. This inversion is not a contingent error; it is the strategy that minimizes any system's local expenditure, and it systematically inverts the four stages of the chain established in VI: elasticity becomes excessive rigidity — R can no longer adapt. Velocity becomes colonizing — R circulates too fast to be stored. Liquidity becomes stratified — accessible to some nodes, nil for others. Utility becomes monolithic — orienting all compositions in the same direction instead of letting the gradient of section IX play out. The two cases below detail the first two; the last two are their direct symmetry, transposed to access and orientation rather than storage and speed.

Excessive rigidity — elasticity becomes so strong that R no longer adapts: it jams instead of storing. Metal-backed currency, whose supply cannot keep up with needs, is the example; Gresham's law is its description: between a high- and a low-density elastic R , agents withdraw the former from circulation.

Fluidity without elasticity — symmetrically, R circulates so fast that it is never stored, and thus, in the precise sense defined in III, ceases to exist as an accumulable residue. Law's system (currency without a conversion constraint) is the historical example; digital attention organized around instant engagement is the contemporary one.

A fifth case is not added to the elasticity-velocity-liquidity-utility chain: it is orthogonal to it, which is what makes it harder to diagnose than the previous four. It is the case where the four conditions *appear* to be satisfied — R seems stored, put into circulation, accessible, orienting — without the condition of uptake posed in II ever having been fulfilled by an actual composition with the term whose trace R claims to carry. Here it is not R that lacks irreversibility; it is the irreversibility that took place elsewhere, in a composition different from the one it claims to attest to, and whose signal is then relocated onto a term that did not contribute to it. This case can be called **capture**, to distinguish it from the internal derangements already described. The criterion that betrays it is not internal to R — from the inside, a captured R has exactly the form of a produced R — it concerns the diversity of the compositions that actually generated it: a genuine R comes from a plurality of encounters independent enough not to be the effect of a single source disguised as several; a captured R can display the same apparent accumulation while coming from a single source repeated under multiple identities. The distinction is therefore not quantitative but structural, and it links directly to the question of underdetermination taken up again in XI: what capture refutes is never the form of R , only a particular R 's claim to have legitimately acquired it.

The five preceding cases are not five independent accidents to be treated one by one; they share a single origin, which must be named so that the section does not close on a mere list. Pathology is never external to the mechanism that produces R — it is not a force coming from outside to disturb an otherwise healthy process. It appears each time this mechanism, the very one that allows a system to change, accumulate, and restart certain cycles, in turn becomes the object of its own exploitation: when the capacity for renewal, necessary to all sustainability in the sense of VII, is turned against the very continuity it is meant to serve. A living system must be able to renew itself; it cannot exploit its own renewal to the point of destroying what gave it its continuity. It is this same origin, and not five disjoint mechanisms, that rigidity, fluidity, and capture each instantiate in their own way.

These derangements combine according to thresholds: a single compromised level leaves the system functional with localized vulnerabilities; all four compromised simultaneously produce total domination, in which the corpus itself (section V) is re-configured to validate the pathology and reproduce itself through ordinary selective transmission.

IX. What R Prescribes: Guidance by Gradient

If directly selecting on R is the error diagnosed in VIII, how does a healthy system orient its future compositions?

Through the orienting property established in III: R does not prescribe a path, it indicates a slope. A prescription imposes an endpoint and sanctions deviation from it; guidance by gradient makes certain compositions easier without forbidding the others, letting actual contingency play its part. The corpus (V) is the material support of this guidance — a differential texture of accumulated R 's, not a code to be deciphered.

This distinction has a direct consequence for correcting a pathology, once it has been diagnosed in VIII: it is not corrected by a constraint added from outside, but by a modification of the slope itself. Forbidding a capture, once it has taken place, is still selecting on a trace — sanctioning an effect rather than touching the conditions that made it possible; and an external constraint remains, by construction, something anyone sophisticated enough can work around by shifting where it applies, as Gresham's law in VIII already shows. Structural correction aims at something else: making the pathological configuration itself less advantageous than ordinary composition, rather than forbidding it alongside it. This is not an absolute guarantee — nothing rules out that a new configuration, unanticipated by the existing gradient, might become advantageous again in turn; this is why guidance by gradient is never a state reached once and for all, but a continuous adjustment, as diachronic as the corpus that carries it.

It is this distinction that explains, in turn, why the pathology of VIII is always an error: the price, the indicator, the signal are not R , they are already partially dissipated traces of it. Selecting on the trace rather than on the conditions is selecting on the effect of an effect while hoping to produce the cause — which never happens.

X. Fourth Return: R as Consciousness

Does the structure of R stop at objects of value, or does it apply to whatever, in us, experiences them?

Section I established that the individual is a stabilized R rather than a primary term. We can now specify what this stabilization produces: consciousness, in the stripped-down sense of a felt continuity of self through change, is the **elastic component of this R** — what, in the flow of mental states, resists dissipation and restores, from one instant to the next, the sensation of being the same. Without this elasticity, each state would be transparent to the next, in the exact sense of section II: a succession without memory, indistinguishable from the absence of relation.

This identity obeys the same sustainability as the four domains of VII: an unsustainable identity- R — broken by trauma — snaps under its own weight; a null identity- R — the dissipation reported in certain states of depersonalization — simply never took hold. A classic question can be directly reformulated in these terms: is a hallucination new content, or the lifting of a filter that had been masking content already present? In this vocabulary: does it belong to the elastic component of R (stored, then released) or to its newly activated viscous component (generated by the dissipation itself)? The data on psychedelics describe a disinhibition, which points toward the filter hypothesis without settling the matter: one observes the lifting of the con-

straint, not what it contained — a particular instance of the general limit section XII will formulate for every R .

XI. What Would Refute R : Objections

Under what conditions would this text be false?

Four objections touch real points, not mere formalities.

The naturalistic fallacy. Does this text slide from a structural observation — R necessarily exists in every non-collapsed system — to a prescription — sustainability ought to be preserved? The answer: the thesis is descriptive of what the word “value” designates in its actual use, not prescriptive. But section VIII, by calling certain configurations pathological, introduces a judgment that goes beyond pure description. This is an open tension, not resolved here.

Underdetermination. The viscoelastic vocabulary (VI) is applied, not deduced — a competing vocabulary could account for the same structures. Mutual information (III) is a demonstrated case, not a general model: that $I(A; B)$ is a computable cross term is a mathematical fact; that economic or affective R 's share the same formal structure remains an analogical extension, not a deduction.

Self-immunizing closure. The system's inability to designate its own originary composition (section XII) could be the sign of an unfalsifiable thesis, where every anomaly becomes confirmation. What would genuinely refute this thesis: a case of a real relation without cost, or of an R transmitted without irreversibility — a counterexample to sections II and III, not a counterargument to the closure itself. The test lies upstream. Capture, described in VIII, specifies where this test must bear in the hard cases: not on the form of R , which is identical whether authentic or captured, but on the actual diversity of the compositions that produced it — a criterion external to R , which keeps the thesis from judging itself on its own terms.

Overextension. Does the extension to consciousness (X) and to the corpus (V) make the framework lose all capacity for exclusion? The partial answer: neither adds a domain of value on the same footing as the four of VII — consciousness applies the structure to the condition of possibility of the experience of value, the corpus to the medium of its transmission. This displaces the difficulty without closing it: how many conditions of possibility can the same scheme legitimately absorb before it stops excluding anything?

The overly convenient isomorphism. Finding, after the fact, that an independent construction — technical, formal, or drawn from an entirely different field — recovers the structure of R could be nothing more than a reading effect: the tendency to see patterns everywhere, including where the parallel is forced. The answer can be no different from the one given in III for mutual information: that an external domain demonstrably shares a formal structure with R is a fact verifiable independently of this text, and it counts for more precisely because the two constructions were not conceived for one another — a text that merely confirms its own premises with its own examples proves nothing, whereas a structure elaborated without it and found

to match it anyway is a fact of a different order. But that this shared structure validates the entire ontological grammar remains, as with III, an analogical extension and not a demonstration. What strengthens the credibility of such a parallel, without ever guaranteeing it, is that the structural properties claimed on each side can each be verified by independent procedures, which do not communicate with each other and which converge nonetheless — a robustness criterion borrowed from any serious experimental method, not an additional philosophical proof. The grammar supplies the form; the external object supplies, at best, invariants that instantiate it without ever fully establishing it.

XII. What R Can Never Say About Itself: Closure

Why this originary composition rather than another?

The ontology establishes that every non-collapsed system carries non-trivial R 's; it does not say which ones, among all those formally possible, actually composed here. Invoking sustainability as a selection criterion would repeat the fundamental error — bringing in a criterion external to composition to decide among possible compositions — since every criterion is itself an R , the product of a prior relation, never a position outside composition.

A consequence of this closure deserves to be singled out, because it is exposed to a confusion symmetrical to the one section I corrects regarding position. R_A , the residue carried by A , needs no access to B to continue accumulating: nothing in its production, as posited by II and III, requires that the encounter with B have already occurred, or even that it be possible in the near term — this is the direct consequence of the temporal indexicality established in I. When A and B finally do compose, their encounter therefore does not retroactively recreate what each had already accumulated separately; that would make the encounter itself the external authority this paragraph has just excluded. The encounter produces new information — an unprecedented $R(A, B)$ in the sense of III, distributed possibly asymmetrically between $R_A(A, B)$ and $R_B(A, B)$ — but it does not replay, and cannot structurally replay, the history already composed on each side before it.

This is where the definition adopted in section III closes on itself out of necessity rather than by default. R is a cross term, not a subtraction: by construction, it has never resided in A alone or in B alone. Seeking the originary composition starting from R is seeking, in the separate terms, something that exists only in their encounter — the same category error as looking for green in blue or in yellow taken in isolation.

If the thesis is true — if everything called value is the residue of an irreversibly accomplished composition — then it is necessary that no system be able to designate its own beginning from within itself: a system that could would contradict the very irreversibility it rests on. Under the reservation maintained in section XI — that this necessity does not exempt the thesis from upstream empirical testing — the impossibility of answering is therefore not R 's failure. It is what R , if it is what this text claims, predicts of itself.

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